

TECHNICAL DOMAIN SPECIFICATION

TicketWave

Business Rules, Financial Algorithms & Domain Architecture Specification

MODULE 01

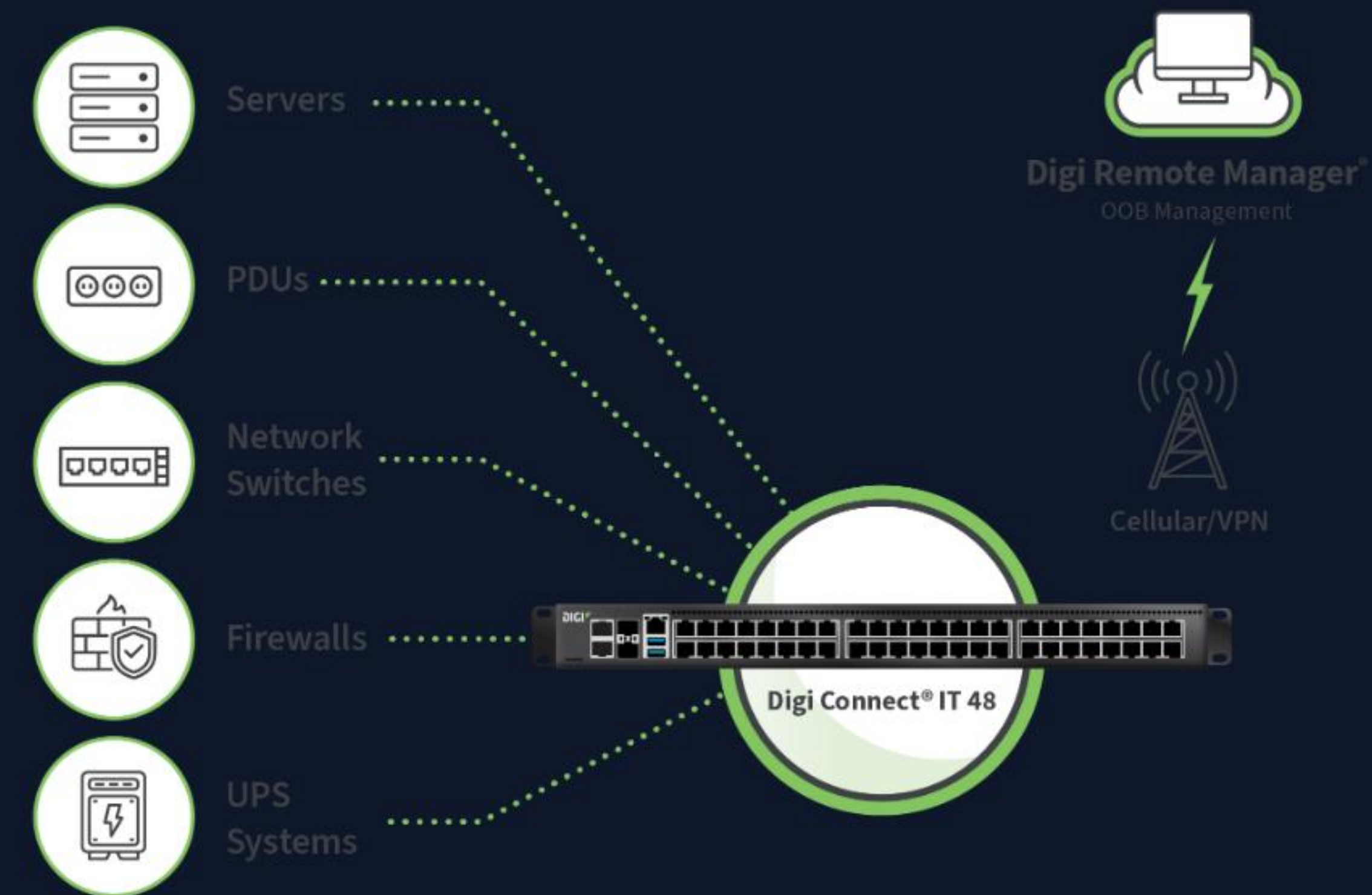
Dynamic Pricing

Additive demand adjustments, protective floor multipliers, and
authoritative promo code rules.

Dynamic Fare Algorithm

$$\text{fare} = \text{baseFare} \times \text{priceModifier} \times \text{demandMultiplier}$$

- ◆ **Rounding:** 2 decimal places using HALF_UP rule.
- ◆ **Demand Multiplier:** $\max(1 + \sum \text{adjustments}, 0.10)$
- ◆ **Protective Floor:** MIN_DEMAND_MULTIPLIER = 0.10 prevents zero or negative fares.
- ◆ **Occupancy Calculation:** Occupied = HELD + BOOKED + BLOCKED + RESERVED_OPERATOR.





Demand Adjustments Reference Table

Adjustment	Trigger Condition	Default Value	Exclusivity Rule
Last-minute	Departure in the future & ≤ 24 h	+0.25	Mutually exclusive with Early-bird
Early-bird	Departure ≥ 30 days in future	-0.10	Mutually exclusive with Last-minute
High-occupancy	Route occupancy $\geq 80\%$	+0.15	Mutually exclusive with Low-occupancy
Low-occupancy	Route occupancy $\leq 20\%$	-0.05	Mutually exclusive with High-occupancy
FareRules	Active rules by route & class	Operator-defined	Stacks additively with all adjustments

Promo Code Validation

Authoritative Application

Method: `applyPromoCode`

Applies a pessimistic row lock (`findByCodeForUpdate`) on the database. Redemption count is incremented with strict concurrency safety.

Display Preview

Method: `previewPromoCode`

Executes a plain read operation without locking (`readOnly`). Usage counter remains untouched while validity & limit checks remain active.

MODULE 02

Seat Holds & Inventory Management

Guaranteed state consistency via Pessimistic Locking, idempotency, and dual-tier expiration handling.

Seat Lifecycle Operations



holdSeat

Assigns seat to purchaser for 10 minutes. Repeat invocation by the same holder refreshes the hold duration (idempotent).



releaseOwnHold

Releases caller's own hold. For non-holders, acts as a silent no-op to maintain state opacity and prevent data leaks.



confirmHold

Converts valid HELD seat to BOOKED. Expired holds throw an exception immediately upon execution.

Booking Lifecycle



MODULE 03

Payments & Refund Policy

Reference-based idempotency, 3D Secure integration, and a two-step settlement workflow.

↻ Cancellation Policy

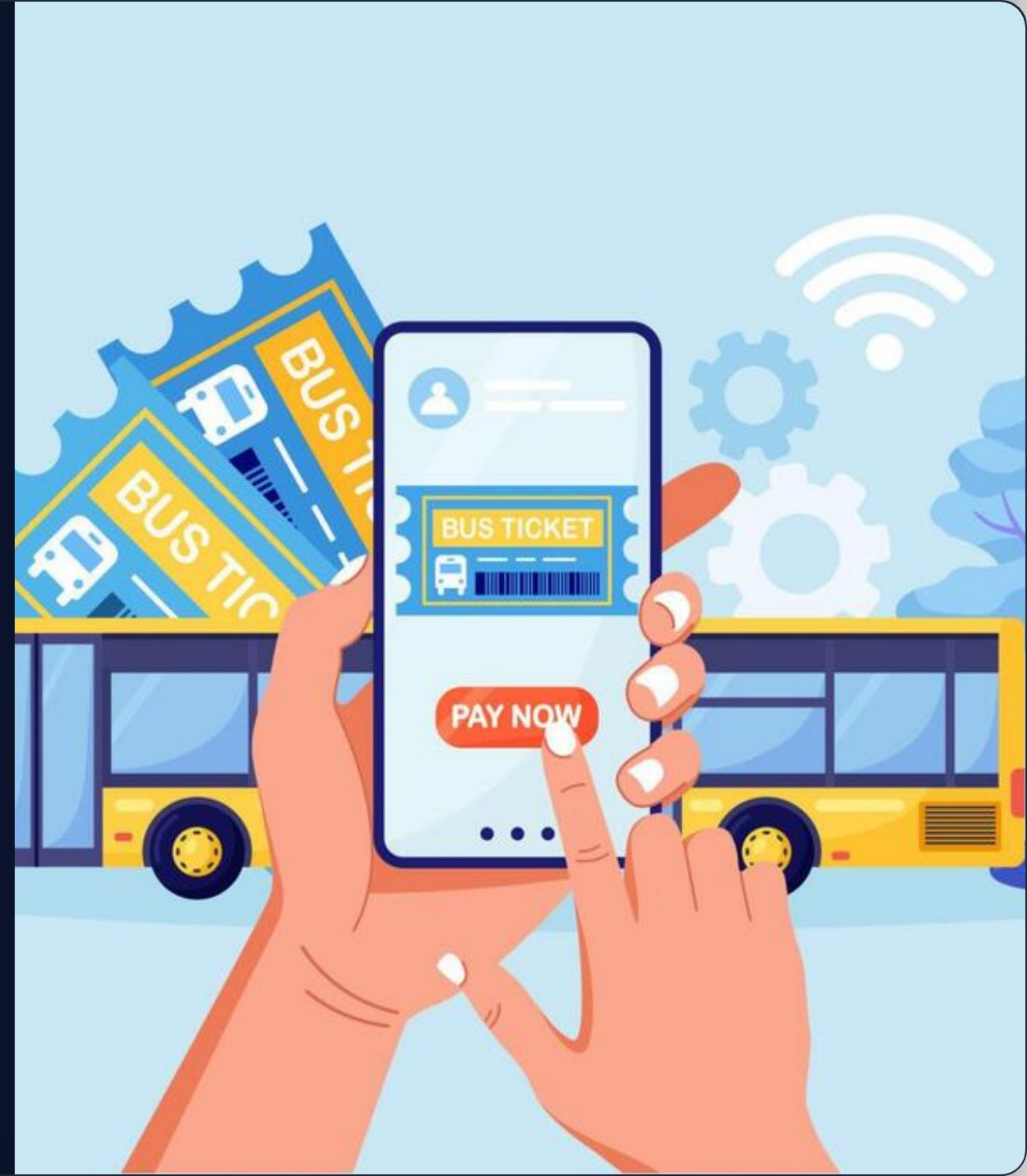
Refund Policy Windows

≥ 7 Days to Departure: 100% full refund (FULL_REFUND).

≥ 24h & < 7 Days: 50% partial refund (PARTIAL_REFUND).

< 24h or Post-Departure: Cancellation strictly blocked by the system.

Note: The exact same time window applies strictly to reschedule requests for confirmed trips.



Two-Step Settlement Model

1. Refund Request

Generates a record with PENDING status.

The booking **remains CONFIRMED** and retains held seats to prevent inventory loss in the event of a subsequent rejection.

2. Operator Decision

Approve: Payment transitions to REFUNDED, booking is cancelled, and seats are released.

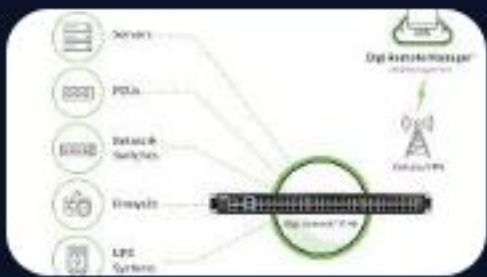
Reject: Keeps trip active and untouched with zero customer impact.

TicketWave Architecture

Domain Validation & Ledger Reconciliation

Active Audit Trail • 100% Line & Branch Test Coverage

Image Sources



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